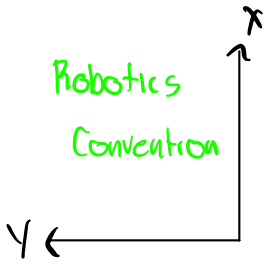
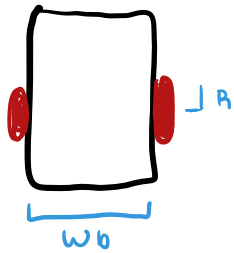


Robot



ΔS = straight forward

$\Delta\theta$ = angle

① Get distance traveled by both wheels using $r \times \Delta\theta$

② Robot $\Delta S = \dfrac{D_left + D_right}{2}$

Simply avg distance travelled

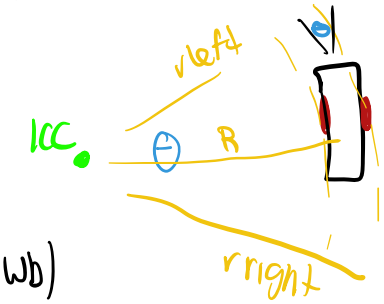
Robot $\Delta\theta = \dfrac{D_left - D_right}{Wb}$

Since

$D_left = \theta \times r_{left}$

$D_right = \theta \times r_{right}$

$= \theta \times (r_{left} + Wb)$



Inst. Center of Curvature

$\rightarrow D_{right} = \theta r_{left} + \theta Wb$

$= D_{left} + \theta Wb$

$\therefore \theta = \dfrac{D_{right} - D_{left}}{Wb}$

Updating Pose

Since whole pose, we use ΔS which is average, and R since that's radius of rotation for whole robot

- $R = \dfrac{disl}{\theta} = \dfrac{\Delta S}{\Delta\theta}$

When No Arc ($\Delta\theta$ < thresh)

- $Y \pm \Delta Y \cdot \sin(\Delta\theta)$
- $X \pm \Delta X \cdot \cos(\Delta\theta)$

When Arc

- $X \pm R\left(\sin(\theta + \Delta\theta) - \sin(\theta)\right)$

  $= X \pm R\sin(\text{new}\theta) - R\sin(\text{old}\,\theta$

- $Y \pm R\left(\cos(\theta + \Delta\theta) - \cos(\theta)\right)$

  $= Y \pm R\cos(\text{new}\theta) - R\cos(\text{old}\,\theta$

Why X with sinθ and Y with cosθ
Because here we are using ICC
with is shifted 90° to our robot's
θ. Sin and cos are shifted 90°

LIDAR

Transforming to World Frame:

reading relative to robot → reading relative to world

1 Shift to curr robot position 2 Rotate by curr angle

- LiDAR rel to robot

$$x_world = x_world + robot_x \cdot \cos(\theta) \quad (y_robot \cdot \sin(\theta)$$

$$y_world = y_world + robot_x \cdot \cos(\theta) \quad (y_robot \cdot \cos(\theta)$$

Matrix:

$$\begin{bmatrix} x_w \\ y_w \\ 1 \end{bmatrix} = \begin{bmatrix} \cos\theta & -\sin\theta & x_r \\ \sin\theta & \cos\theta & y_r \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x_l \\ y_l \\ 1 \end{bmatrix}$$